# Aniruddh Ballal

Software engineer building full-stack and mobile applications, with a background in scientific computing.

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## EXPERIENCE

**Indiassetz** | Assistant Software Engineer — Mar - Oct 2026

- Built an investor/back-office platform with Next.js, FastAPI and PostgreSQL: role-based document approvals, capital-call settlement tracking and portfolio reporting.
- Migrated inherited React Native apps across framework and native-build changes; repaired authentication, navigation and third-party integrations.
- Added CRM payment-review and exception-approval workflows, revenue-attribution dashboards and spreadsheet exports.

**Qwixpert Consulting** | Data Analyst Intern — Jan - Mar 2026

- Analysed client datasets, automated reporting across linked Excel workbooks and prepared consulting presentations.
- Independently built supply-chain network design, demand-forecast comparison and dispatch-planning tools with maps, dashboards and spreadsheet reports.

**Indiassetz** | Frontend Web Development Intern — Jun - Jul 2025

- Built React/Material UI interfaces and reusable components, refining interactions and layouts with design and engineering teams.

**IISc, Department of Physics** | Solar Physics Intern — Jun - Aug 2024

- Modelled solar magnetic-field evolution using Python, spherical harmonics, NumPy and SciPy under Dr. Prantika Bhowmik; visualised solar-cycle behaviour.

## SELECTED PROJECTS

**Sunalyse** Source — 2025-26
Reconstructed coronal fields across 181 Carrington rotations with a Python PFSS pipeline and Three.js viewer. Vectorisation and four-core tracing achieved a documented **7.6x speedup** over the original implementation.

**Queen's Gambit 2.0** Source — 2025-26
Built file steganography using legal chess moves, Rust/WASM and Web Workers, with optional AES-256-GCM encryption. Documented 32.1 KB encoding benchmark: **195 s to 5.2 s**.

**dedgram** Source — 2026
Built an Instagram research toolkit with Android capture, Playwright scraping and a public atlas; stored profile/reel snapshots, source evidence and username history to investigate suspected automated accounts.

**kNupdate** Source — 2026
Built a Kotlin/Compose activity-sharing app with paired-device delivery and SQLite histories. A Cloudflare relay discards activity payloads after all recipients acknowledge or 30 seconds elapse.

**SatLies** Source — 2026
Implemented a Three.js orbital simulator with RK4 integration, comparing Earth's oblateness (J2) and lunar/solar gravity against a two-body baseline through live trajectories and separation plots.

**Credex** Source — 2026
Built a Next.js/Node.js trade-finance prototype linking procurement, payment records and loan review; a Python/FastAPI service uses PuLP to schedule payments under cash-flow and deadline constraints.

## TECHNICAL SKILLS

**Languages:** TypeScript/JavaScript, Python, Kotlin, Rust, SQL.
**Applications & data:** React, Next.js, React Native, Node.js/Express, FastAPI, PostgreSQL, MongoDB, SQLite.
**Scientific & tooling:** Three.js, NumPy, SciPy, WebAssembly, Playwright, Git, Docker.

## EDUCATION

**Manipal Institute of Technology** — Graduated Jul 2026
B.Tech, Information Technology | CGPA: 8.1/10

## LEADERSHIP & ENGINEERING ACTIVITIES

**ThrustMIT** - Avionics member; PCB, telemetry and RF/GPS/IMU integration for student rocketry (2022-23).
**Manipal Runners' Club** - Co-organised the 2024 marathon (8,000+ participants) and supported six-week runner training.